

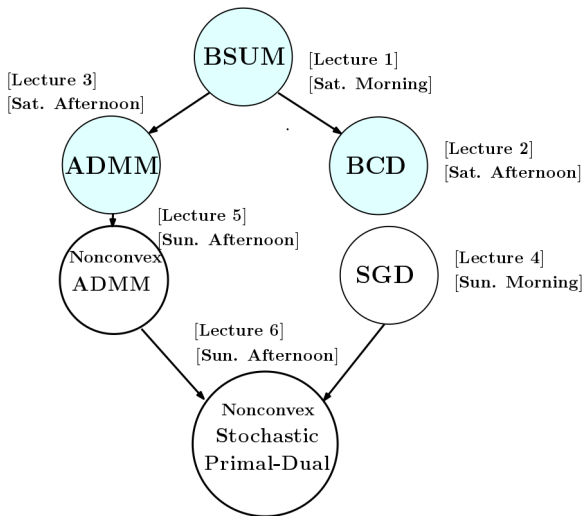
Lecture 6: Optimizing Nonconvex Finite Sums: The NESTTS Algorithm

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Schedule (temporary)



Reference

- D. Hajinezhad, M. Hong, T. Zhao and Z. Wang, "NESTT: A Nonconvex Primal-Dual Splitting Method for Distributed and Stochastic Optimization", arXiv:1605.07747, 2016

Outline

- 1 Introduction
 - Motivation
 - Related Works
 - Applications
- 2 The Proposed Approach
- 3 NESTT-G Algorithm
 - NESTT-G Details
 - Convergence Analysis of NESTT-G
- 4 NESTT-E Algorithm
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- 5 Connections with Existing Works
- 6 Numerical Results

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Problem Formulation

- Consider:

$$\min_{z \in Z} f(z) := \frac{1}{N} \sum_{i=1}^N g_i(z) + g_0(z) + p(z), \quad (1)$$

- g_i , $i = 1, \dots, N$ are cost functions; g_0 and p are regularizers.
- Z : Close and Convex set.
- g_i : L_i -smooth, possibly **nonconvex**.
- $p(z)$: lower semi-continuous, **convex**, possibly **nonsmooth**.

Motivation

- When N is very large, would like to optimize one function at a time
- Basically we would like to **decompose across the functions**
- But all existing fast algorithms only work for **convex** functions
- Can we find **fast algorithms** to achieve decomposition for non-convex problems?
- Can we rigorously characterize the convergence rates?

This work

- Studying the problem from a **primal-dual** perspective.
- We develop a stochastic algorithm that achieves **sublinear** convergence rates; constants can be up to $\mathcal{O}(N)$ times better than the gradient method.
- For special quadratic problems, achieve **linear** convergence.
- Identified some interesting connection between the proposed primal-dual approach, with a few primal only methods such as **IAG/SAG/SAGA**.

Related Works in Convex Case

- Extensive literature in convex case.
- Gradient Descent (GD):

$$x^{r+1} = x^r - \alpha \frac{1}{N} \sum_{i=1}^N \nabla g_i(x_i^r)$$

- Stochastic Gradient Descent (SG) [Robbins and Monro, 1951]:

Randomly pick an index $i_r \in \{1, \dots, N\}$

$$x^{r+1} = x^r - \frac{1}{\eta_{i_r}} \nabla g_{i_r}(x^r)$$

Related Works in Convex Case

- Stochastic Average Gradient (SAG) [Le Roux 12]:
- Consider a special case of problem (1) : $p(x) = 0, g_0(x) = 0$

Randomly pick an index $i_r \in \{1, \dots, N\}$

$$y_{i_r}^r = x^r, \quad y_j^r = y_j^{r-1}, \quad \forall j \neq i_r$$

$$\begin{aligned} x^{r+1} &= x^r - \frac{1}{\eta N} \sum_{i=1}^N \nabla g_i(y_i^r) \\ &= x^r - \frac{1}{\eta N} \sum_{i=1}^N \nabla g_i(y_i^{r-1}) + \frac{1}{\eta N} (\nabla g_{i_r}(y_{i_r}^{r-1}) - \nabla g_{i_r}(x^r)) \end{aligned}$$

- We only need to keep track of the sum $\sum_{i=1}^N \nabla g_i(y_i^{k-1})$, as well as each of its components.

Related Works in Convex Case

- The SAGA algorithm [Defazio et al 14]:
- Suppose $g_0(x) = 0$ in problem (1).

Randomly pick an index $i_k \in \{1, \dots, N\}$

$$y_{i_k}^k = x^k, \quad y_j^k = y_j^{k-1}, \quad \forall j \neq i_k$$

$$w^{k+1} = x^k - \frac{1}{\eta N} \sum_{i=1}^N \nabla g_i(y_i^{k-1}) + \frac{1}{\eta} (\nabla g_{i_k}(y_{i_k}^{k-1}) - \nabla g_{i_k}(x^k))$$

$$x^{k+1} = \text{prox}_h^{1/\eta}(w^{k+1}) \quad (\text{Deal with nonsmooth part})$$

- It has better convergence rates.
- Requires storage of past N gradients.

Related Works in Convex Case

- Other related methods :
 - MISO [Mairal, 2013].
 - SDCA [Shalev-Schwartz and Zhang, 2013].
- But, these all introduce memory requirements.
- Is it possible to achieve similar convergence rates as SAG, but without additional storage?
- **Yes!** Stochastic variance reduced gradient (SVRG) proposed in [Johnson-Zhang 13].

Related Works in Convex Case

- The iteration of SVRG given by

Iterate: for $s = 1, 2, \dots$

$$\tilde{x} = \tilde{x}_{s-1}, \quad \tilde{z} = \frac{1}{N} \sum_{i=1}^N \nabla g_i(\tilde{x}), \quad x_0 = \tilde{x}$$

Iterate: for $t = 1, 2, \dots$

Randomly pick i_t

$$x_t = x_{t-1} - \frac{1}{\eta} (\nabla g_{i_t}(x_{t-1}) - \nabla g_{i_t}(\tilde{x}) + \tilde{z})$$

End

Option I: set $\tilde{x}_s = x_m$

Option II: set $\tilde{x}_s = x_t$ for randomly chosen $t \in \{0, \dots, m-1\}$

End

Related Works in Nonconvex Case

- Classical algorithms
 - Incremental Methods [Bertsekas 11]
 - SGD based algorithms [Ghadimi- Lan 13]
- Recent fast algorithms
 - Nonconvex SVRG [Zhu et al 16, Reddi et al 16]
 - Non-convex SAGA [Reddi et al 16]
- The latter two algorithms only deal with smooth cases

Application 1: High Dimensional Regression

- $y \in \mathbb{R}^M$ observation vector generated: $y = X\nu + \epsilon$
 - $X \in \mathbb{R}^{M \times P}$: covariate matrix.
 - $\nu \in \mathbb{R}^P$: ground truth.
 - $\epsilon \in \mathbb{R}^M$: noise.
- $A = X + W$ where $W \in \mathbb{R}^{M \times P}$ is the noise.
- **Goal**: To estimate the ground truth ν .
- Set $\hat{\Gamma} := 1/M(A^\top A) - \Sigma_W$, and $\hat{\gamma} := 1/M(A^\top y)$.
- **Non-Convex** Optimization [Loh, Wainwright 12]:

$$\min_z z^\top \hat{\Gamma} z - \hat{\gamma} z \quad \text{s.t.} \quad \|z\|_1 \leq R.$$

Application 2: Sparse Principal Component Analysis

- Formulation [Richtarik 12]:

$$\max_x \|Dx\|_2^2 - \lambda r(x), \quad \text{s.t. } \|x\|_2^2 \leq 1$$

- $\|Dx\|_2^2$ represents the explained variance through x .
- $r(x)$ is sparsity-promoting, and $\lambda > 0$ controls the sparsity
- $D \in \mathbb{R}^{n \times p}$ is the data matrix
- n represents the total number of documents
- p represents the total words

Application 2: Sparse Principal Component Analysis

- Splitting the **rows** of $D \in \mathbb{R}^{n \times p}$ into N sub-matrix; each row represents **a few documents**

$D =$

D_1			
D_2			
...	...	...	...
D_N			

- SPCA problem can be reformulated:

$$\max_x \sum_{i=1}^N \|D_i x\|_2^2 - \lambda r(x), \quad \text{s.t. } \|x\|_2^2 \leq 1. \quad (2)$$

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A Reformulation

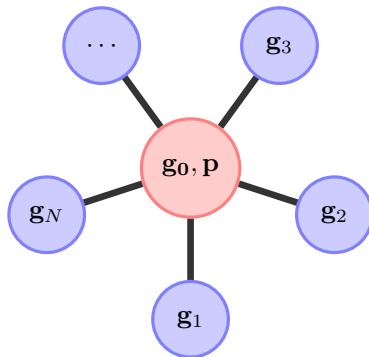
- Motivated by the distributed nonconvex ADMM [H.-Razaviyayn-Luo 14]
- Let us split z into N new variables
- consider the following reformulation:

$$\min_{x, z \in \mathbb{R}^d} \frac{1}{N} \sum_{i=1}^N g_i(x_i) + g_0(z) + h(z), \text{ s.t. } x_i = z, i = 1, \dots, N \quad (3)$$

where $h(z) := \iota_Z(z) + p(z)$, $x := [x_1; \dots; x_N]$.

A Reformulation

- Distributed Setting:



Preliminary

- Augmented Lagrangian:

$$L(x, z; \lambda) = \sum_{i=1}^N \left(\frac{1}{N} g_i(x_i) + \langle \lambda_i, x_i - z \rangle + \frac{\eta_i}{2} \|x_i - z\|^2 \right) + g_0(z) + h(z)$$

- $\lambda := \{\lambda_i\}_{i=1}^N$ is the set of dual variables.
- $\eta := \{\eta_i > 0\}_{i=1}^N$ are penalty parameters.
- For simplicity, define: $g(z) = \sum_{i=1}^N \frac{1}{N} g_i(z)$

Assumptions

A-(a) $f(z)$ is bounded from below, i.e. $\underline{f} := \min_{z \in Z} f(z) > -\infty$.

A-(b) The g_i 's and g have Lipschitz continuous gradients, i.e.,

$$\begin{aligned}\|\nabla g_i(y) - \nabla g_i(z)\| &\leq L_i \|y - z\|, \quad \forall y, z \in \text{dom}(g_i), \quad i = 0, \dots, N \\ \|\nabla g(y) - \nabla g(z)\| &\leq L \|y - z\|, \quad \forall y, z \in \text{dom}(g).\end{aligned}$$

A-(c) $\eta_i > L_i/N$. If g_0 is nonconvex, then $\sum_{i=1}^N \eta_i > 3L_0$.

- From assumption A-(b): $L \leq 1/N \sum_{i=1}^N L_i$, and the equality can be achieved in the worst case.
- From Assumption A-(c): $L(x, z; \lambda)$ is strongly convex w.r.t. each x_i and z , with modulus $\gamma_i := \eta_i - L_i/N$ and $\gamma_z = \sum_{i=1}^N \eta_i - L_0$, respectively.

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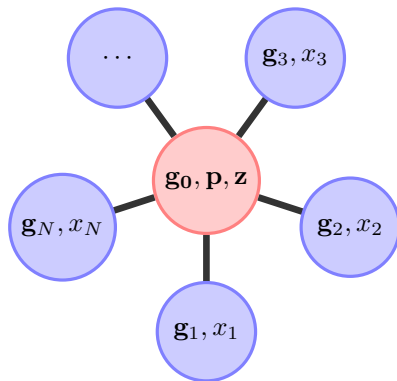
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- From assumption A-(b): $L \leq 1/N \sum_{i=1}^N L_i$, and the equality can be achieved in the worst case.
- From Assumption A-(c): $L(x, z; \lambda)$ is **strongly convex** w.r.t. each x_i and z , with modulus $\gamma_i := \eta_i - L_i/N$ and $\gamma_z = \sum_{i=1}^N \eta_i - L_0$, respectively.

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Overview of The NESTT-G Algorithm

- Consider Distributed Setting:



- Central node** is able to **broadcast** to every node, and each node is able to send messages to the central node.

NESTT-G Algorithm

- Consider $L(x, z, \lambda)$:

$$L(x, z; \lambda) = \sum_{i=1}^N \left(\frac{1}{N} g_i(x_i) + \langle \lambda_i, x_i - z \rangle + \frac{\eta_i}{2} \|x_i - z\|^2 \right) + g_0(z) + h(z)$$

- Let us define:

$$V_i(x_i, z; \lambda_i) = \underbrace{\frac{1}{N} g_i(z) + \frac{1}{N} \langle \nabla g_i(z), x_i - z \rangle}_{\text{an approximation of } \frac{1}{N} g_i(x_i)} + \langle \lambda_i, x_i - z \rangle + \frac{\alpha_i \eta_i}{2} \|x_i - z\|^2.$$

- Remarks:

- 1 We have approximated $\frac{1}{N} g_i(x_i)$ use its linear approximation
- 2 Parameter α_i gives some freedom to the algorithm.

NESTT-G Algorithm

The NESTT-G Algorithm

From iteration $r = 1, \dots, R$, do:

Pick $i(r) \in \{1, 2, \dots, N\}$ with probability $p_{i(r)}$ and update (x_j, λ_j) :

$$x_{i(r)}^{r+1} = \arg \min_{x_{i(r)}} V_{i(r)} \left(x_{i(r)}, z^r, \lambda_{i(r)}^r \right);$$

$$\lambda_{i(r)}^{r+1} = \lambda_{i(r)}^r + \alpha_{i(r)} \eta_{i(r)} \left(x_{i(r)}^{r+1} - z^r \right);$$

$$\lambda_j^{r+1} = \lambda_j^r, \quad x_j^{r+1} = z^r, \quad \forall j \neq i(r);$$

Update z :
$$z^{r+1} = \arg \min_{z \in Z} L(\{x_i^{r+1}\}, z; \lambda^r).$$

Output: (z^m, x^m, λ^m) where m randomly picked from $\{1, 2, \dots, R\}$.

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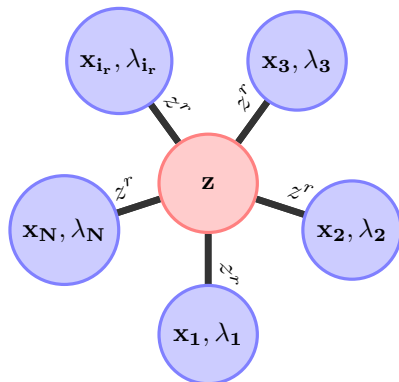
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Output: (z^m, x^m, λ^m) where m randomly picked from $\{1, 2, \dots, R\}$.

NSETT-G Algorithm

- First, each node i stores (x_i, λ_i)
- When the algorithm starts, the central node **broadcast** to everyone z^r

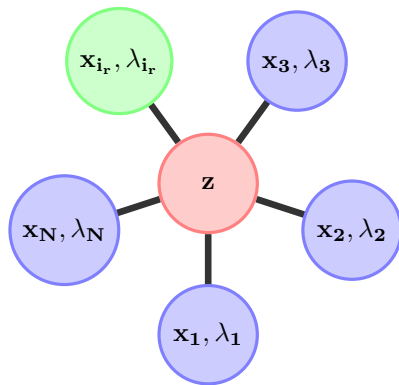


NSETT-G Algorithm

- Randomly pick $i(r)$ and update $x_{i(r)}$, and related $\lambda_{i(r)}$:

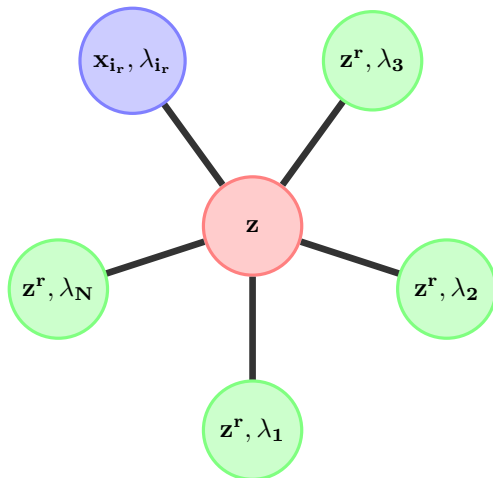
$$x_{i(r)}^{r+1} = \arg \min_{x_{i(r)}} V_{i(r)}(x_{i(r)}, z^r, \lambda_{i(r)}^r);$$

$$\lambda_{i(r)}^{r+1} = \lambda_{i(r)}^r + \alpha_{i(r)} \eta_{i(r)} (x_{i(r)}^{r+1} - z^r)$$



NSETT-G Algorithm

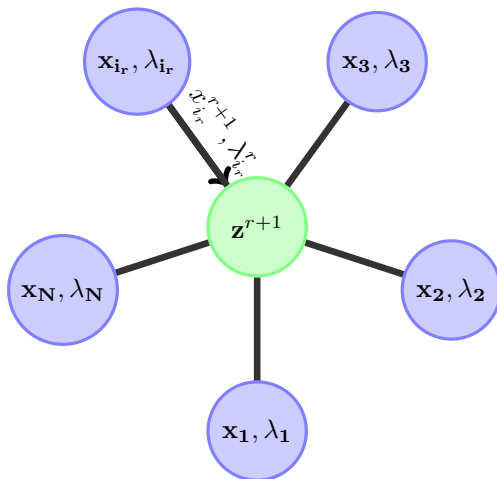
- When $j \neq i(r)$ simply set $x_j^{r+1} = z^r$, and no change on the dual variable $\lambda_j^{r+1} = \lambda_j^r$



NSETT-G Algorithm

- Note i_r transmits to node z ; Update z :

$$z^{r+1} = \arg \min_{z \in Z} L(\{x_i^{r+1}\}, z; \lambda^r)$$



Definitions

- $r(j)$: the **last iteration** in which the j th block is picked before iteration $r + 1$

$$r(j) := \max\{t \mid t < r + 1, j = i(t)\}.$$

- $y_j^r := z^{r(j)}$ if $j \neq i(r)$, and $y_{i(r)}^r = z^r$.
- $\mathcal{F}^r$: σ -field generated by $\{(z^t, x^t)\}_{t=1}^r$.

Key Properties of NESTT-G

- Optimality condition of the x , by the x_{i_r} update rule, we obtain

$$x_{i(r)}^{r+1} = z^r - \frac{1}{\alpha_{i(r)}\eta_{i(r)}} \left(\lambda_{i(r)}^r + \frac{1}{N} \nabla g_{i(r)}(z^r) \right), \quad (4)$$

- By the fact that λ_j^r is updated we have [to be shown in the next slide]

$$x_j^{r+1} = z^r = z^r - \underbrace{\frac{1}{\alpha_j\eta_j} (\lambda_j^r + \frac{1}{N} \nabla g_j(z^{r(j)}))}_{=0}, \quad \forall j \neq i(r). \quad (5)$$

Key Properties of NESTT-G

- Update scheme of the λ :

$$\begin{aligned}\lambda_{i(r)}^{r+1} &= -\frac{1}{N} \nabla g_{i(r)}(z^r), \\ \lambda_j^{r+1} &= -\frac{1}{N} \nabla g_j(z^{r(j)}), \quad \forall j \neq i(r). \quad \text{[why?]} \end{aligned}$$

- Unified form of dual update:

$$\lambda_i^{r+1} = -\frac{1}{N} \nabla g_i(y_i^r) \tag{6}$$

- **Observation:** The dual variables are remembers the past gradients of g_i 's.

Main Steps of Convergence Analysis of NESTT-G

- **Step 1)** Build a **Potential Function** and show that it is decreasing?

$$Q^r := \sum_{i=1}^N \frac{1}{N} g_i(z^r) + g_0(z^r) + h(z^r) + \sum_{i=1}^N \frac{3p_i\eta_i}{(\alpha_i\eta_i)^2} \|1/N(\nabla g_i(y_i^{r-1}) - \nabla g_i(z^r))\|^2$$

- **Step 2)** Show that certain optimality gap is decreasing in a sublinear manner. The optimality gap is defined as:

$$\mathbb{E}[G^r] := \frac{1}{\beta^2} \mathbb{E} \left[\|z^r - \text{prox}_h^{1/\beta} [z^r - \beta \nabla (g(z^r) + g_0(z^r))] \|^2 \right],$$

where $\beta = 1 / \sum_{i=1}^N \eta_i$ [does this potential function make sense?]

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where $\beta = 1 / \sum_{i=1}^N \eta_i$ **[does this potential function make sense?]**

The Potential Function

- What does the potential function represent?
- The **Potential Function** is an upper bound of $L(x, y, \lambda)$
- To see this, first note, for all $j \neq i(r)$

$$\frac{1}{N}g_j(x_j^{r+1}) + \langle \lambda_j^r, x_j^{r+1} - z^r \rangle + \frac{\eta_j}{2} \|x_j^{r+1} - z^r\|^2 = \frac{1}{N}g_j(z^r)$$

- Also note, for the index that is picked we have

$$\begin{aligned} & \frac{1}{N}g_{i(r)}(x_{i(r)}^{r+1}) + \langle \lambda_{i(r)}^r, x_{i(r)}^{r+1} - z^r \rangle + \frac{\eta_i}{2} \|x_{i(r)}^{r+1} - z^r\|^2 \\ & \stackrel{D.L.}{\leq} \frac{1}{N}g_{i(r)}(z^r) + \frac{\eta_{i(r)} + L_{i(r)}/N}{2} \|x_{i(r)}^{r+1} - z^r\|^2 \\ & = \frac{1}{N}g_{i(r)}(z^r) + \frac{\eta_{i(r)} + L_{i(r)}/N}{2(\alpha_{i(r)}\eta_{i(r)})^2} \|1/N(\nabla g_{i(r)}(y_{i(r)}^{r-1}) - \nabla g_{i(r)}(z^r))\|^2 \end{aligned}$$

the last inequality is due to the optimality condition of $x_{i(r)}^{r+1}$

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$$\frac{1}{N}g_j(x_j^{r+1}) + \langle \lambda_j^r, x_j^{r+1} - z^r \rangle + \frac{\eta_j}{2} \|x_j^{r+1} - z^r\|^2 = \frac{1}{N}g_j(z^r)$$

- Also note, for the index that is picked we have

$$\begin{aligned} & \frac{1}{N}g_{i(r)}(x_{i(r)}^{r+1}) + \langle \lambda_{i(r)}^r, x_{i(r)}^{r+1} - z^r \rangle + \frac{\eta_i}{2} \|x_{i(r)}^{r+1} - z^r\|^2 \\ & \stackrel{D.L.}{\leq} \frac{1}{N}g_{i(r)}(z^r) + \frac{\eta_{i(r)} + L_{i(r)}/N}{2} \|x_{i(r)}^{r+1} - z^r\|^2 \\ & = \frac{1}{N}g_{i(r)}(z^r) + \frac{\eta_{i(r)} + L_{i(r)}/N}{2(\alpha_{i(r)}\eta_{i(r)})^2} \|1/N(\nabla g_{i(r)}(y_{i(r)}^{r-1}) - \nabla g_{i(r)}(z^r))\|^2 \end{aligned}$$

the last inequality is due to the optimality condition of $x_{i(r)}^{r+1}$

The Potential Function

- What does the potential function represent?
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the last inequality is due to the optimality condition of $x_{i(r)}^{r+1}$

The Potential Function

- Take expectation on augmented Lagrangian we have:

$$\begin{aligned}
 & \mathbb{E}_{x^{r+1}}[L(x^{r+1}, z^r; \lambda^r) \mid \mathcal{F}^r] \\
 & \leq \sum_{i=1}^N \frac{1}{N} g_i(z^r) + \sum_{i=1}^N \frac{p_i(\eta_i + L_i/N)}{2(\alpha_i \eta_i)^2} \|1/N(\nabla g_i(y_i^{r-1}) - \nabla g_i(z^r))\|^2 \\
 & \quad + g_0(z^r) + h(z^r) \\
 & \leq \sum_{i=1}^N \frac{1}{N} g_i(z^r) + \sum_{i=1}^N \frac{3p_i \eta_i}{(\alpha_i \eta_i)^2} \|1/N(\nabla g_i(y_i^{r-1}) - \nabla g_i(z^r))\|^2 \\
 & \quad + g_0(z^r) + h(z^r) = Q^r \rightarrow \text{Potential Function}
 \end{aligned}$$

- The expectation is taken on the random selection of $i(r)$ (each with probability p_i)
- The last inequality is due to the assumption that $\eta_i > L_i/N$

Convergence Analysis of NESTT-G

- Sufficient Descent:

Lemma

Suppose Assumption A holds, and pick

$$\alpha_i = p_i = \beta \eta_i, \text{ where } \beta := \frac{1}{\sum_{i=1}^N \eta_i}, \text{ and } \eta_i \geq \frac{24L_i}{Np_i} \quad (7)$$

Then the following descent estimate holds true for NESTT-G

$$\begin{aligned} \mathbb{E}[Q^r - Q^{r-1} | \mathcal{F}^{r-1}] &\leq -\frac{\sum_{i=1}^N \eta_i}{8} \mathbb{E}_{z^r} \|z^r - z^{r-1}\|^2 \\ &\quad - \sum_{i=1}^N \frac{1}{\eta_i} \left\| \frac{1}{N} (\nabla g_i(z^{r-1}) - \nabla g_i(y_i^{r-2})) \right\|^2. \end{aligned} \quad (8)$$

Comment

- Due to the randomness, here we are bounding the **expected** descent
- The penalty parameter η_i has to be large enough; similar as the nonconvex ADMM case
- The probability is **proportional to the penalty parameter** η_i
- The descent is measured by

$$\mathbb{E}_{z^r} \|z^r - z^{r-1}\|^2 \quad \text{and} \quad \left\| \frac{1}{N} (\nabla g_i(z^{r-1}) - \nabla g_i(y_i^{r-2})) \right\|^2 \quad (9)$$

Proof of Lemma

- **Step 1).** Using the definition of potential function Q^r :

$$\begin{aligned}
 & \mathbb{E}[Q^r - Q^{r-1} \mid \mathcal{F}^{r-1}] \\
 &= \mathbb{E} \left[\sum_{i=1}^N \frac{1}{N} (g_i(z^r) - g_i(z^{r-1})) + g_0(z^r) - g_0(z^{r-1}) + h(z^r) - h(z^{r-1}) \right] \\
 &+ \mathbb{E} \sum_{i=1}^N \frac{3p_i}{\alpha_i^2 \eta_i} \left\| \frac{1}{N} \nabla g_i(z^r) - \frac{1}{N} \nabla g_i(y_i^{r-1}) \right\|^2 \\
 &- \sum_{i=1}^N \frac{3p_i}{\alpha_i^2 \eta_i} \left\| \frac{1}{N} \nabla g_i(z^{r-1}) - \frac{1}{N} \nabla g_i(y_i^{r-2}) \right\|^2. \tag{10}
 \end{aligned}$$

- The expectation is taking over z^r

Proof of Lemma

- **Step 2).** Bound the **Blue** term:

$$\begin{aligned}
 & \mathbb{E} \left[\sum_{i=1}^N \frac{1}{N} (g_i(z^r) - g_i(z^{r-1})) + g_0(z^r) - g_0(z^{r-1}) + h(z^r) - h(z^{r-1}) \right] \\
 & \leq \sum_{i=1}^N \frac{\beta}{\alpha_i} \left\| \frac{1}{N} \nabla g_i(z^{r-1}) - \frac{1}{N} \nabla g_i(y_i^{r-2}) \right\|^2 \\
 & \quad - \left(\frac{3}{4\beta} - \frac{\sum_{i=1}^N L_i/N + 3L_0}{2} \right) \mathbb{E}_{z^r} \|z^r - z^{r-1}\|^2.
 \end{aligned}$$

Proof of Lemma

- **Step 3)** Bound the Red term:

$$\begin{aligned}
 & \mathbb{E} \sum_{i=1}^N \frac{3p_i}{\alpha_i^2 \eta_i} \left\| \frac{1}{N} \nabla g_i(z^r) - \frac{1}{N} \nabla g_i(y_i^{r-1}) \right\|^2 \\
 & - \frac{3p_i}{\alpha_i^2 \eta_i} \left\| \frac{1}{N} \nabla g_i(z^{r-1}) - \frac{1}{N} \nabla g_i(y_i^{r-2}) \right\|^2 \\
 & \leq \sum_{i=1}^N \frac{3L_i^2}{\alpha_i \eta_i N^2} (p_i + (1-p_i)(1+\xi_i)) \mathbb{E}_{z^r} \|z^r - z^{r-1}\|^2 \\
 & + \frac{3}{\alpha_i \eta_i} \left((1-p_i)(1+\frac{1}{\xi_i}) - 1 \right) \left\| \frac{1}{N} \nabla g_i(y_i^{r-2}) - \frac{1}{N} \nabla g_i(z^{r-1}) \right\|^2.
 \end{aligned}$$

- **Step 4)** Combining Step 2 and Step 3, take $\xi = 3/p_i$: Done!

Convergence Analysis of NESTT-G

- Sublinear Convergence of NESTT-G:

Definition

The *proximal gradient* of problem (1) is given by (for any $\gamma > 0$)

$$\begin{aligned}\tilde{\nabla} f_{\gamma}(z) &:= \gamma \left(z - \text{prox}_{p+\iota_Z}^{\gamma} [z - 1/\gamma \nabla(g(z) + g_0(z))] \right) \\ \text{where } \text{prox}_{p+\iota_Z}^{\gamma} [u] &:= \underset{u \in Z}{\operatorname{argmin}} \quad p(u) + \frac{\gamma}{2} \|z - u\|^2.\end{aligned}$$

- Optimality Gap:

$$\begin{aligned}\mathbb{E}[G^r] &:= \mathbb{E} \left[\|\tilde{\nabla}_{1/\beta} f(z^r)\|^2 \right] \\ &= \frac{1}{\beta^2} \mathbb{E} \left[\|z^r - \text{prox}_h^{1/\beta} [z^r - \beta \nabla(g(z^r) + g_0(z^r))] \|^2 \right].\end{aligned}$$

Convergence Analysis of NESTT-G

Theorem (Main Theorem)

Suppose Assumption A holds, and pick (for $i = 1, \dots, N$)

$$\alpha_i = p_i = \frac{\sqrt{L_i/N}}{\sum_{j=1}^N \sqrt{L_j/N}}, \quad \eta_i = 4 \left(\sum_{j=1}^N \sqrt{L_j/N} \right) \sqrt{L_i/N}, \quad \beta = \frac{1}{4(\sum_{i=1}^N \sqrt{L_i/N})^2}.$$

1) Every limit point of NESTT-G is stationary solution of problem (3) w.p. 1.

$$2) \mathbb{E}[G^m] \leq 48 \left(\sum_{i=1}^N \sqrt{L_i/N} \right)^2 \frac{\mathbb{E}[Q^0 - Q^R]}{R};$$

$$3) \mathbb{E}[G^m] + \mathbb{E} \left[\sum_{i=1}^N 3\eta_i^2 \|x_i^m - z^{m-1}\|^2 \right] \leq 48 \left(\sum_{i=1}^N \sqrt{L_i/N} \right)^2 \frac{\mathbb{E}[Q^0 - Q^R]}{R}.$$

Proof of Theorem

- Part 1) From previous lemma:

- There exists $c > 0$ such that:

$$\begin{aligned}\mathbb{E}[Q^r | \mathcal{F}^{r-1}] &\leq Q^{r-1} - c\mathbb{E}_{z^r} \|z^r - z^{r-1}\|^2 \\ &\quad - \sum_{i=1}^N \frac{1}{\eta_i} \|\nabla g_i(z^{r-1}) - \nabla g_i(y_i^{r-2})\|^2\end{aligned}$$

- $f(z)$ is lower bounded, so $\{Q^r\}$ is a bounded sequence (Lower Bound: $\underline{Q}$):
 $\{Q^r - \underline{Q}\} \geq 0$.

- Apply **Supermartingale Convergence** theorem [Bertsekas 97] $\{Q^r\}$ converges a.s. and:

$$\|\nabla g_i(z^{r-1}) - \nabla g_i(y_i^{r-2})\|^2 \rightarrow 0, \quad \mathbb{E}_{z^r} \|z^r - z^{r-1}\| \rightarrow 0, \quad \text{a.s.}$$

Proof of Theorem

- Using $\|\nabla g_i(z^{r-1}) - \nabla g_i(y_i^{r-2})\|^2 \rightarrow 0$ we conclude

$$\|\lambda_{i(r)}^r - \lambda_{i(r)}^{r-1}\| \rightarrow 0$$

- Using the dual update equation: $\|x_{i(r)}^r - z^{r-1}\| \rightarrow 0$ which further implies:

$$\|z^r - z^{r-1}\| \rightarrow 0$$

- Finally we can conclude:

$$\|x_i^r - x_i^{r-1}\| \rightarrow 0, \quad \|\lambda_i^r - \lambda_i^{r-1}\| \rightarrow 0, \quad \text{a.s.,} \quad \forall i.$$

- Successive differences of all the primal and dual variables go to zero a.s.

Proof of Theorem

- Part 2) We bound the gap:

$$\begin{aligned} & \mathbb{E} \left[\|z^r - \text{prox}_h^{1/\beta}[z^r - \beta \nabla(g(z^r) + g_0(z^r))]\|^2 \right] \\ & \leq 6\mathbb{E}\|z^r - z^{r-1}\|^2 + 3 \sum_{i=1}^N \frac{\beta}{\eta_i} \mathbb{E} \left\| \frac{1}{N} \nabla g_i(z^{r-1}) - \frac{1}{N} \nabla g_i(y_i^{r-2}) \right\|^2 \end{aligned}$$

- If we set $\eta_i = \frac{24}{p_i N}$, we get:

$$\alpha_i = p_i = \frac{\sqrt{L_i/N}}{\sum_{j=1}^N \sqrt{L_j/N}}, \quad \eta_i = \sqrt{24L_i/N} \sum_{j=1}^N \sqrt{24L_j/N}$$

Proof of Theorem (cont)

- Using the definition of G^r :

$$\begin{aligned}\mathbb{E}[G^r] &\leq \frac{6}{\beta^2} \mathbb{E} \|z^r - z^{r-1}\|^2 + \sum_{i=1}^N \frac{3}{\beta \eta_i} \mathbb{E} \left\| \frac{1}{N} \nabla g_i(z^{r-1}) - \frac{1}{N} \nabla g_i(y_i^{r-2}) \right\|^2 \\ &\stackrel{(8)}{\leq} \frac{48}{\beta} \mathbb{E}[Q^{r-1} - Q^r] = 48 \left(\sum_{i=1}^N \sqrt{L_i/N} \right)^2 \mathbb{E}[Q^{r-1} - Q^r].\end{aligned}$$

- Summing both side and use the definition of z^m

$$\mathbb{E}[G^m] \leq 48 \left(\sum_{i=1}^N \sqrt{L_i/N} \right)^2 \frac{\mathbb{E}[Q^0 - Q^R]}{R}$$

Linear Convergence of NESTT-G

- Assumption B:

B-(a) Each function $g_i(z)$ is a quadratic function of the form

$$g_i(z) = 1/2 z^T A_i z + \langle b, z \rangle$$

where A_i is a symmetric matrix but not necessarily positive semidefinite;

B-(b) The feasible set Z is a closed compact polyhedral set;

B-(c) The nonsmooth function $p(z) = \mu \|z\|_1$, for some $\mu \geq 0$.

Linear Convergence of NESTT-G

Lemma (Luo-Tseng 92)

Suppose Assumptions A and B hold. Let Z^* denotes the set of stationary solutions of problem (1), and $\text{dist}(z, Z^*) := \min_{u \in Z^*} \|z - u\|$. Then we have the following

- ① **(Error Bound Condition)** For any $\xi \geq \min_z f(z)$, exists a positive scalar τ such that the following error bound holds

$$\text{dist}(z, Z^*) \leq \tau \|\tilde{\nabla}_{1/\beta} f(z)\| \quad (11)$$

for all $z \in (Z \cap \text{dom } h)$ and $z \in \{z : f(z) \leq \xi\}$.

- ② **(Separation of Isocost Surfaces)** There exists a scalar $\delta > 0$ such that

$$\|z - v\| \geq \delta \quad \text{whenever} \quad z \in Z^*, v \in Z^*, f(z) \neq f(v). \quad (12)$$

Linear Convergence of NESTT-G

Theorem (Linear Convergence of NESTT-G)

Suppose that Assumptions A, B are satisfied. Then the sequence $\{\mathbb{E}[Q^{r+1}]\}_{r=1}^{\infty}$ converges Q -linearly to some $Q^ = f(z^*)$, where z^* is a stationary solution for problem (1). That is, there exists a finite $\bar{r} > 0$, $\rho \in (0, 1)$ such that for all $r \geq \bar{r}$,*

$$\mathbb{E}[Q^r - Q^*] < \rho \mathbb{E}[Q^{r+1} - Q^*]$$

Proof of Theorem

- (x^r, z^r, λ^r) converges to (x^*, z^*, λ^*) . Therefore,

$$Q^* = f(z^*) = \sum_{i=1}^N 1/N g_i(z^*) + g_0(z^*) + p(z^*).$$

Then $\exists \bar{r} > 0$ and $\bar{v} \in \mathbb{R}$ s.t. $\forall r > \bar{r}$

$$f(\bar{z}^r) = \bar{v}, \quad \forall r \geq \bar{r}$$

where $\bar{z}^r = \arg \min_{z \in Z^*} \|z^r - z\|$.

- We know $\|x^{r+1} - x^r\| \rightarrow 0$, $\|z^{r+1} - z^r\| \rightarrow 0$, $\|x_i^{r+1} - z^{r+1}\| \rightarrow 0$, $\|\lambda^{r+1} - \lambda^r\| \rightarrow 0$, so

$$L(\bar{z}^r, \bar{x}^r, \bar{\lambda}^r) = f(\bar{z}^r) = \bar{v}, \quad \forall r \geq \bar{r},$$

- Now we prove that the expectation of $\Delta^{r+1} := Q^{r+1} - \bar{v}$ diminishes **Q-linearly**.

Proof of Theorem (cont)

- **Step 1:** There exists $\sigma_1 > 0$ such that

$$\mathbb{E}[Q^r - Q^{r+1}] \geq \sigma_1 \left(\mathbb{E}\|z^{r+1} - z^r\|^2 + \sum_{i=1}^N \mathbb{E}\|1/N \nabla g_i(z^r) - 1/N \nabla g_i(y_i^{r-1})\|^2 \right);$$

- **Step 2:** There exists $\tau > 0$ such that:

$$\mathbb{E}\|z^r - \bar{z}^r\|^2 \leq \tau \|\mathbb{E}[\nabla_{1/\beta} \tilde{f}(z^r)]\|^2;$$

- **Step 3:** There exists $\sigma_2 > 0$ such that:

$$\|\mathbb{E}[\nabla_{1/\beta} \tilde{f}(z^r)]\|^2 \leq \sigma_2 \left(\mathbb{E}\|z^{r+1} - z^r\|^2 + 1/N \sum_{i=1}^N \mathbb{E}\|\nabla g_i(z^r) - \nabla g_i(y_i^{r-1})\|^2 \right);$$

- **Step 4:** There exists $\sigma_3 > 0$ such that:

$$\begin{aligned} \mathbb{E}[Q^{r+1} - \bar{v}] &\leq \sigma_3 (\mathbb{E}\|z^r - \bar{z}^r\|^2 + \mathbb{E}\|z^{r+1} - z^r\|^2 \\ &\quad + \sum_{i=1}^N \mathbb{E}\|1/N \nabla g_i(z^r) - 1/N \nabla g_i(y_i^{r-1})\|^2). \end{aligned}$$

Proof of Theorem (cont)

- Combining step 4 and step 2: for $r \geq \bar{r}$

$$\begin{aligned} \mathbb{E}[Q^{r+1} - \bar{v}] &\leq \sigma_3(\tau \|\mathbb{E}[\nabla_{1/\beta} \tilde{f}(z^{r-1})]\|^2 + \mathbb{E}\|z^{r+1} - z^r\|^2 \\ &\quad + \sum_{i=1}^N \mathbb{E}\|1/N \nabla g_i(z^r) - 1/N \nabla g_i(y_i^{r-1})\|^2). \end{aligned}$$

- Using step 3:

$$\mathbb{E}[Q^{r+1} - \bar{v}] \leq \sigma_4 \left(\mathbb{E}\|z^{r+1} - z^r\|^2 + \sum_{i=1}^N \mathbb{E}\|1/N \nabla g_i(z^r) - 1/N \nabla g_i(y_i^{r-1})\|^2 \right).$$

- Using step 1:

$$\mathbb{E}[Q^{r+1} - \bar{v}] \leq \frac{\sigma_4}{\sigma_1} \mathbb{E}[Q^r - Q^{r+1}], \quad \forall r \geq \bar{r},$$

So, for $\sigma_5 = \frac{\sigma_4}{\sigma_1} > 0$, we have

$$\mathbb{E}[\Delta^{r+1}] \leq \frac{\sigma_5}{1 + \sigma_5} \mathbb{E}[\Delta^r], \quad \forall r \geq \bar{r}.$$

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Overview of NESTT-E Algorithm

- **Motivation:**

- 1 In NESTT-G every agent should update its local variable at every iteration, not possible in practice! (broadcasting not available)
- 2 Performing exact minimization instead of taking a gradient step can speed up the algorithms in distributed setting.

- **Algorithm Setup:** Let us define

$$\begin{aligned} U(x, z; \lambda) &= \sum_{i=1}^N U_i(x_i, z; \lambda_i) \\ &= \sum_{i=1}^N \left(\frac{1}{N} g_i(x_i) + \langle \lambda_i, x_i - z \rangle + \frac{\alpha_i \eta_i}{2} \|x_i - z\|^2 \right) \\ &\quad + p(z) + g_0(z). \end{aligned}$$

- Note that if $\alpha_i = 1$ for all i , then the $U(x, z; \lambda) = L(x, z; \lambda)$.

NESTT-E Algorithm

The NESTT-E Algorithm

From iteration $r = 1, \dots, R$, do:

$$z^{r+1} = \arg \min_z L(x^r, z; \lambda^r). \quad (13)$$

Randomly pick $i(r) \in \{1, 2, \dots, N\}$ with probability $p_{i(r)}$:

$$x_{i(r)}^{r+1} = \underset{x_{i(r)}}{\operatorname{argmin}} U_{i(r)}(x_{i(r)}, z^{r+1}; \lambda_{i(r)}^r); \quad (14)$$

$$\lambda_{i(r)}^{r+1} = \lambda_{i(r)}^r + \alpha_{i(r)} \eta_{i(r)} \left(x_{i(r)}^{r+1} - z^{r+1} \right); \quad (15)$$

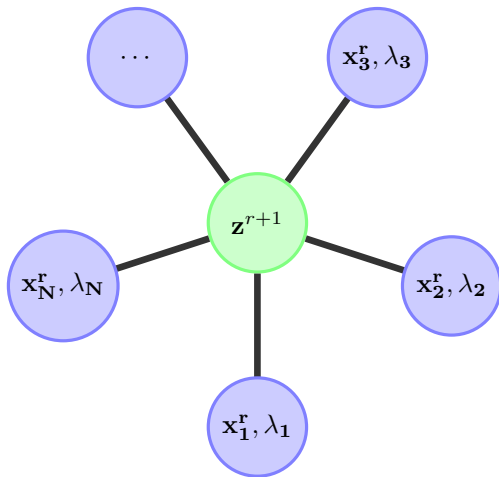
$$x_j^{r+1} = x_j^r, \quad \lambda_j^{r+1} = \lambda_j^r \quad \forall j \neq i(r). \quad (16)$$

Output: (z^m, x^m, λ^m) where m is randomly picked from $\{1, 2, \dots, R\}$.

NSETT-E Algorithm

- Update z :

$$z^{r+1} = \arg \min_{z \in Z} L(\{x_i^r\}, z; \lambda^r)$$

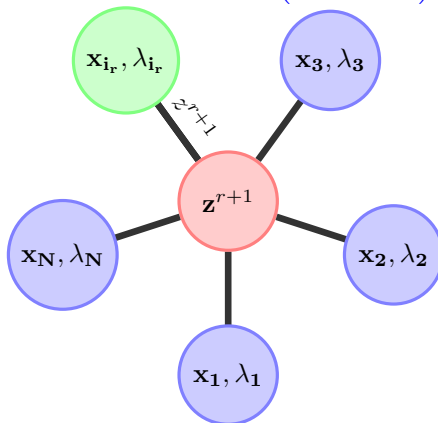


NSETT-G Algorithm

- Randomly pick $i(r)$ and update $x_{i(r)}$, and related $\lambda_{i(r)}$:

$$x_{i(r)}^{r+1} = \arg \min_{x_{i(r)}} V_{i(r)}(x_{i(r)}, z^{r+1}, \lambda_{i(r)}^r);$$

$$\lambda_{i(r)}^{r+1} = \lambda_{i(r)}^r + \alpha_{i(r)} \eta_{i(r)} (x_{i(r)}^{r+1} - z^{r+1})$$



Some Key Properties of NESTT-E

- for $i = i(r)$ optimality condition of x_i :

$$\frac{1}{N} \nabla g_{i(r)}(x_{i(r)}^{r+1}) + \lambda_{i(r)}^r + \alpha_{i(r)} \eta_{i(r)} (x_{i(r)}^{r+1} - z^{r+1}) = 0. \quad (17)$$

- For due variable:

$$\frac{1}{N} \nabla g_{i(r)}(x_{i(r)}^{r+1}) = -\lambda_{i(r)}^{r+1}. \quad (18)$$

- The optimality condition for the z -update:

$$z^{r+1} = \text{prox}_h \left[z^{r+1} - \nabla_z (L(x^r, z, \lambda^r) - h(z)) \right] \quad (19)$$

$$= \text{prox}_h \left[z^{r+1} - \sum_{i=1}^N \eta_i \left(z^{r+1} - x_i^r - \frac{\lambda_i^r}{\eta_i} \right) - \nabla g_0(z^{r+1}) \right]. \quad (20)$$

Convergence Analysis of NESTT-E

- For simplicity:

$$w := (x, z, \lambda), \quad \beta := \frac{1}{\sum_{i=1}^N \eta_i}, \quad c_i := \frac{L_i^2}{\alpha_i \eta_i N^2} - \frac{\gamma_i}{2} + \frac{1 - \alpha_i}{\alpha_i} \frac{L_i}{N}, \quad \alpha := \{\alpha_i\}_{i=1}^N.$$

$$\hat{x}_i^{r+1} := \arg \min_{x_i} U_i(x_i, z^{r+1}, \lambda_i^r), \quad \hat{\lambda}_i^{r+1} := \lambda_i^r + \alpha_i \eta_i (\hat{x}_i^{r+1} - z^{r+1}).$$

- Prox-Gradient of $L(x, z, \lambda)$:

$$\tilde{\nabla} L(w) = [z - \text{prox}_h[z - \nabla_z(L(w) - h(z))]; \nabla_{x_1} L(w); \dots \nabla_{x_N} L(w)].$$

- Optimality Gap:

$$H(w^r) := \|\tilde{\nabla} L(w^r)\|^2 + \sum_{i=1}^N \frac{L_i^2}{N^2} \|x_i^r - z^r\|^2.$$

Sublinear Convergence of NESTT-E

Theorem (Sublinear Convergence of NESTT-E)

Suppose Assumption A holds, (η_i, α_i) are chosen such that $c_i < 0$. Then

1) Potential Function $[L(x, z, \lambda)]$, is decreasing and lower bounded:

$$\mathbb{E}[L(w^r)] \geq \mathbb{E}[L(w^{r+1})] \geq \underline{f} > -\infty, \quad \forall r \geq 0.$$

2) Almost surely every limit point of $\{w^r\}$ is a stationary solution of (3).

3) For some function of α denoted as $C(\alpha)$, we have the following:

$$\mathbb{E}[H(w^m)] \leq \frac{C(\alpha)\mathbb{E}[L(w^1) - L(w^{R+1})]}{R}. \quad (21)$$

Proof of Theorem

- Proof of Part 1, 2:
- **Step 1)** For every $r \geq 1$ we can prove that:

$$\mathbb{E}[L^{r+1} - L^r | x^r, z^r] \leq -\frac{\gamma z}{2} \|z^{r+1} - z^r\|^2 + \sum_{i=1}^N p_i c_i \|x_i^r - \hat{x}_i^{r+1}\|^2.$$

- **Step 2)** We can show that L^r is lower bounded, i.e. $L^r \geq \underline{f}$

Proof of Theorem (Part 3)

- **Step 1)** There exists $\sigma_1(\alpha) > 0$:

$$H(w^r) \leq \sigma_1(\alpha) \left(\|z^r - z^{r+1}\|^2 + \sum_{i=1}^N \|x_i^r - \hat{x}_i^{r+1}\|^2 \right).$$

- **Step 2)** : There exists $\sigma_2(\alpha) > 0$:

$$E[L^r - L^{r+1}|F^r] \geq \sigma_2(\alpha) \left(\sum_{i=1}^N \|\hat{x}_i^{r+1} - x_i^r\|^2 + \|z^{r+1} - z^r\|^2 \right).$$

- **Step 3)** Step 1 + Step 2:

$$H(w^r) \leq \frac{\sigma_1(\alpha)}{\sigma_2(\alpha)} E[L^r - L^{r+1}|F^r].$$

- **Step 4)** : Summing both side, set $C(\alpha) = \frac{\sigma_1(\alpha)}{\sigma_2(\alpha)}$ and use the definition of $w^m = (x^m, z^m, \lambda^m)$:

$$\mathbb{E}[H(w^m)] \leq \frac{C(\alpha) \mathbb{E}[L(w^1) - w^{R+1}]}{R}.$$

Linear Convergence of NESTT-E

Theorem (Linear Convergence of NESTT-E)

Suppose that Assumptions A, B are satisfied. Then the sequence $\{\mathbb{E}[L^{r+1}]\}_{r=1}^{\infty}$ converges *Q-linearly* to some $L^* = f(z^*)$, where z^* is a stationary solution for problem (1). That is, there exists a finite $\bar{r} > 0$, $\rho \in (0, 1)$ such that for all $r \geq \bar{r}$,

$$\mathbb{E}[L^r - L^*] \leq \rho \mathbb{E}[L^{r+1} - L^*]$$

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Connections and Comparisons with Existing Works

Proposition

The NESTT-G can be written into the following compact form:

$$z^{r+1} = \arg \min_z h(z) + g_0(z) + \frac{1}{2\beta} \|z - u^{r+1}\|^2$$

$$u^{r+1} := z^r - \beta \left(\frac{1}{N\alpha_{i(r)}} (\nabla g_{i(r)}(z^r) - \nabla g_{i(r)}(y_{i(r)}^{r-1})) - \frac{1}{N} \sum_{i=1}^N \nabla g_i(y_i^{r-1}) \right)$$

	$h(z)$	$g_0(z)$	α_i	p_i
IAG [Blatt 07]	0	0	1	cyclic
SAG [Schmidt 12]	0	0	1	1/N
SAGA [Defazio 14]	✓	0	1/N	1/N

Comparison in terms of number of gradient evaluations

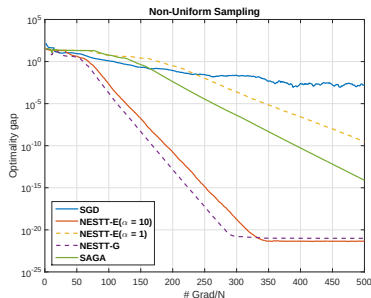
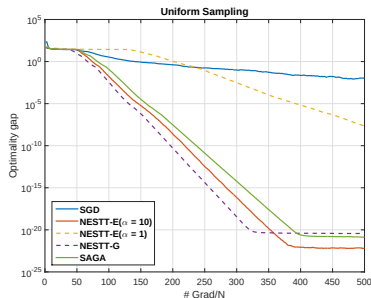
- Based on the classical theory of GD to get ϵ - stationary solution GD should be run in the order of $\mathcal{O}(L/\epsilon)$.
- Number of gradient evaluations in each step fo GD is N . So in the worse case ($\sum_{i=1}^N L_i/N = L$) number of gradient evaluations is $\mathcal{O}\left(\sum_{i=1}^N L_i/\epsilon\right)$.

	NESTT-G	GD
# of Gradient	$\mathcal{O}\left((\sum_{i=1}^N \sqrt{L_i/N})^2/\epsilon\right)$	$\mathcal{O}\left(\sum_{i=1}^N L_i/\epsilon\right)$
Case I: $L_i = 1, \forall i$	$\mathcal{O}(N/\epsilon)$	$\mathcal{O}(N/\epsilon)$
Case II : $\mathcal{O}(N^{2/3})$: $L_i = N^{2/3}$ rest: $L_i = 1$	$\mathcal{O}(N/\epsilon)$	$\mathcal{O}(N^{4/3}/\epsilon)$
Case III : $\mathcal{O}(\sqrt{N})$: $L_i = N$ rest: $L_i = 1$	$\mathcal{O}(N/\epsilon)$	$\mathcal{O}(N^{3/2}/\epsilon)$
Case IV : $\mathcal{O}(1)$: $L_i = N^2$ rest: $L_i = 1$	$\mathcal{O}(N/\epsilon)$	$\mathcal{O}(N^2/\epsilon)$

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High Dimensional Regression

- Comparison of NESTT-G/E, SAGA, SGD. The x -axis denotes the number of passes of the dataset.
- Left: Uniform Sampling ($N_i = N_j$); Right: Non-uniform Sampling ($N_i \neq N_j$).



High Dimensional Regression

- Optimality gap $\|\tilde{\nabla}_{1/\beta} f(z^r)\|^2$ for different algorithms, with 100 passes of the datasets.

N	SGD		NESTT-E ($\alpha = 10$)		NESTT-G		SAGA	
	Uniform	Non-Uni	Uniform	Non-Uni	Uniform	Non-Uni	Uniform	Non-Uni
10	3.4054	0.2265	2.6E-16	6.16E-19	2.3E-21	6.1E-24	2.7E-17	2.8022
20	0.6370	6.9087	2.4E-9	5.9E-9	1.2E-10	2.9E-11	7.7E-7	11.3435
30	0.2260	0.1639	3.2E-6	2.7E-6	4.5E-7	1.4E-7	2.5E-5	0.1253
40	0.0574	0.3193	5.8E-4	8.1E-5	1.8E-5	3.1E-5	4.1E-5	0.7385
50	0.0154	0.0409	8.3E-4	7.1E-4	1.2E-4	2.7E-4	2.5E-4	3.3187

- In both uniform and non-uniform case NESTT-G outperforms other methods.

Conclusion:

- Presented Primal-Dual based algorithms (NESTT).
- Proved the Sublinear Convergence in general, and Q-Linear convergence for certain nonconvex ℓ_1 penalized quadratic problems.
- Popular primal only methods such as IAG/SAG/SAGA are special cases of NESTT.
- Showed that NESTT-G with *non-uniform sampling* achieves an ϵ -stationary solution of (1) using $\mathcal{O}((\sum_{i=1}^N \sqrt{L_i/N})^2/\epsilon)$ gradient evaluations, which can be up to $\mathcal{O}(N)$ times better than GD where the L_i 's are not uniform.

Future Works:

- **Potential Research Direction:**
- Find a tighter bound for the step size.
- Reveal connections between NESTT and other primal only methods such as SVRG.